

Tranquilizers

By Yitagesu T(MSc.)
2024

Tranquilizers

- Give **strong sedation** without producing sleep and produce a state of **indifference** and **disinterest**.
- Effective in reducing excitation, agitation, aggressiveness and impulsiveness (major tranquilizers)
- They may be classified into
 - **Major tranquilizers:** used mainly for treatment of psychosis (schizophrenia and mania)
 - **Minor tranquilizers:** used to reduce pathological anxiety, agitation and tension

Tranquilizers...

- **Dopamine hypothesis** of Schizophrenia (and mania) suggests that schizophrenia results from increased **dopaminergic neurotransmission**
- Hence the strategy is to block these dopaminergic receptors (specifically D2 and D3 receptors).

Dopaminergic...

Three dopaminergic pathways

The mesolimbic pathway

- The mesolimbic pathway is important for **memory** and for **motivating behaviours**.
- By blocking this pathway, antipsychotic drugs reduce the intense emotions caused by conditions such as schizophrenia.

Dopaminergic pathways

The mesocortical pathway

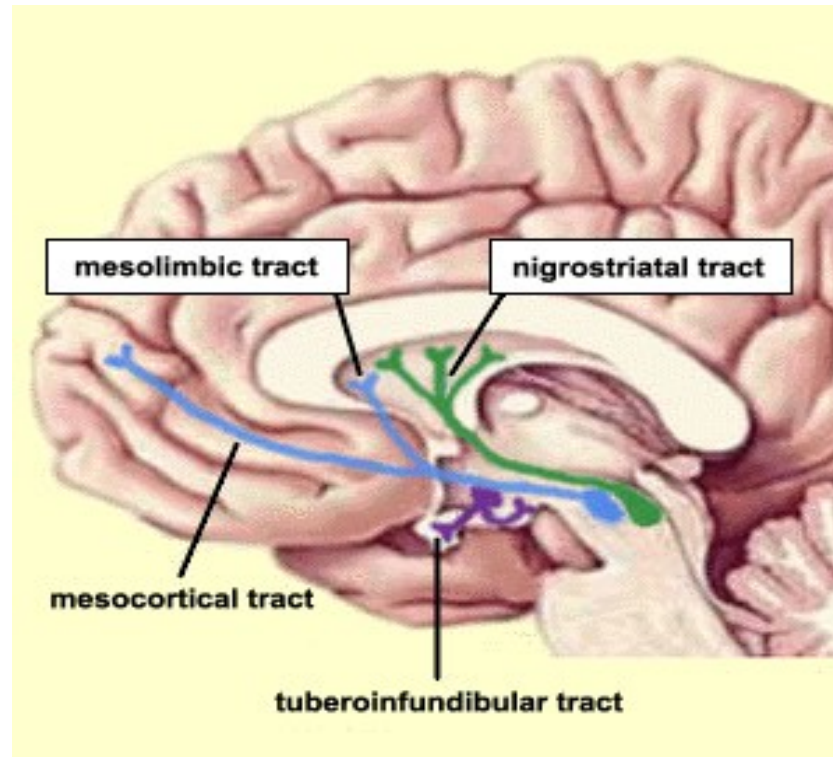
- Some evidence indicates that a malfunction in this pathway might be the cause of some of the symptoms of **schizophrenia**, such as **hallucinations** and **disordered thinking**.
- Medications that block this pathway reduce psychotic delirium, but also reduce the overall activity of the frontal lobes.

Dopaminergic...

The nigrostriatal pathway

- involved in motor control
- Degeneration of the neurons in this pathway is associated with the **trembling** and **muscular rigidity** symptomatic of Parkinson's disease.

Dopaminergic...



Major neuroleptics

- **Major neuroleptics:**
 1. Tricyclic neuroleptics
 2. The fluorobutyrophenone derivatives

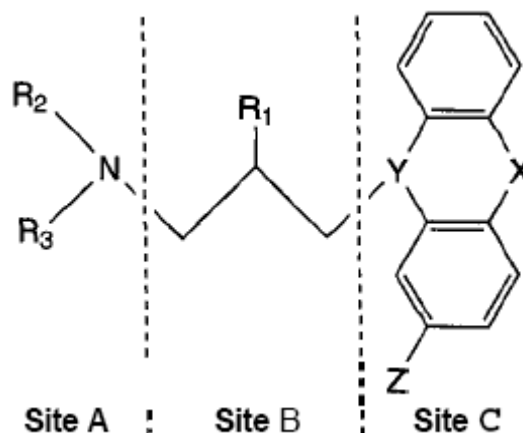
Major neuroleptics...

1. Tricyclic Neuroleptics

- After the introduction of chlorpromazine (CPZ) as a treatment for schizophrenia, thousands of new agents based on its tricyclic topology were prepared and examined pharmacologically.
- The neuroleptic potential of these agents was determined by measuring its "chlorpromazine index".

Major neuroleptics...

- The tricyclic antipsychotics may be dissected into three substructures as shown below,
 - those being the pendent amine functionality (Site A),
 - the diaryl heterotricyclic (Site C), and
 - the intervening alkyl chain (Site B)

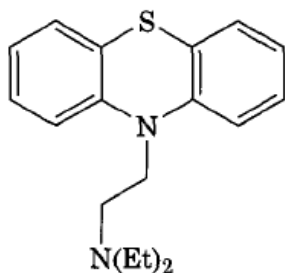


Major neuroleptics...

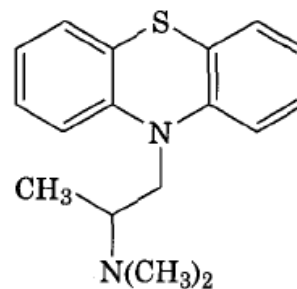
SAR

1. Modification at site B

- The distance between Sites A and C is critical for neuroleptic activity, with a **three carbon** chain being optimal.
- Shortening the chain to **two carbons** has the effect of amplifying the **anticholinergic** and **antihistaminic properties**.



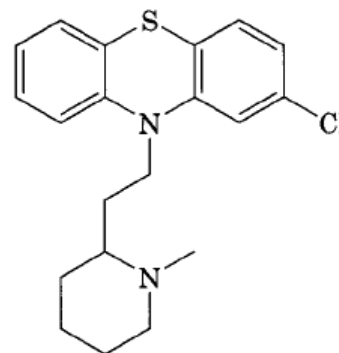
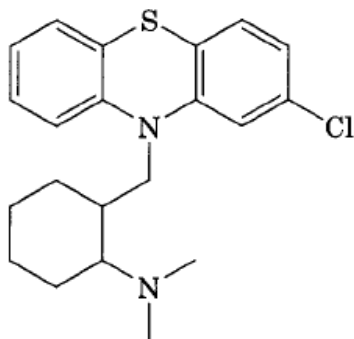
diethazine



promethazine

Major neuroleptics...

- small alkyl substituents such as methyl are tolerated at the **C2** carbon, larger substituents (i.e., R = phenyl) that restrict the free rotation decrease neuroleptic potency.
- Additionally, **if rotation is restricted through ring formation, the chlorpromazine** index is greatly reduced



thioridazine

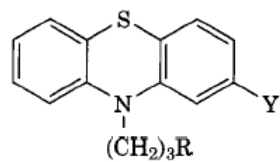
Major neuroleptics...

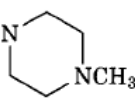
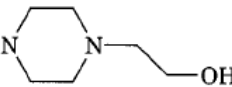
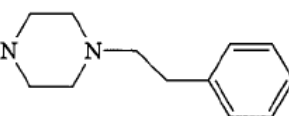
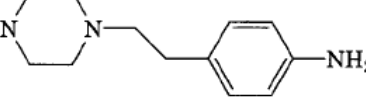
- **Effect of Aromatic Substitution within the Tricyclic System**

- substitution at position 2 provides compounds of enhanced potency in blocking conditioned response in rats
- The electronic nature of the substituent also plays a role in determining the efficacy in this model.
- Electron withdrawing groups such as **chloro** and **trifluoromethyl** show superior conditioned response blocking activity relative to the corresponding **alkyl** or **alkoxy** derivatives

Major neuroleptics...

- **Nature of the Amino Group**
 - The size and nature of the basic amino group has considerable influence on the behavioral profile of the phenothiazine neuroleptics
 - The effect is shown below

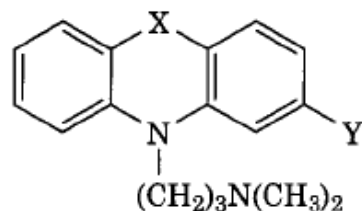


R	X	CPZ Index Blockade of Conditioned Response in Rats
$\text{N}(\text{Et})_2$	Cl	0.8
	Cl	0.7
	Cl	0.2
	Cl	0.05
	Cl	2.7
	Cl	9.0
	CF_3	3.0
	CF_3	9.0
	Cl	

Major neuroleptics...

- Variations within the Tricyclic Topology**

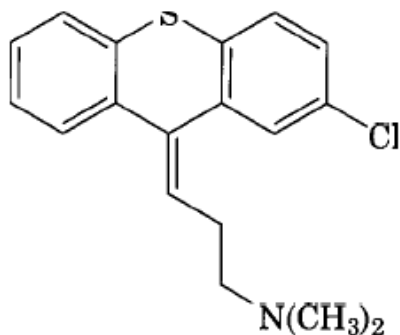
- Introduction of other group VI elements in place of the sulfur in the **phenothiazines** produced the corresponding **phenoxazine** and **phenoselenazine** derivatives



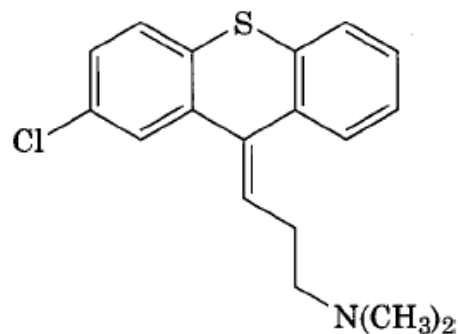
X	Y	Chlorpromazine Index Blockade of Conditioned Response in Rats
S	Cl	1
S	CF ₃	1.7
O	Cl	0.38
O	CF ₃	0.29
Se	Cl	0.1
Se	CF ₃	1.1
CH ₂	Cl	~0.1
C(CH ₃) ₂	Cl	>0.1

Major neuroleptics...

- A unique series of neuroleptics results from the replacement of the nitrogen within the phenothiazine ring system with a **methane** carbon. These analogues are known as **thioxanthenes**
- The introduction of the double bond within the propylamino chain provides for geometric isomers (*cis* and *trans*)



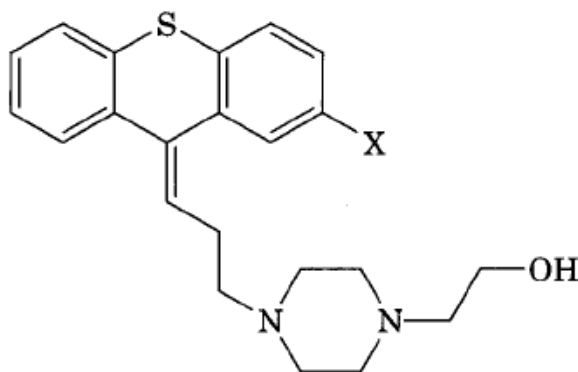
Cis chlorprothixine



trans chlorprothixine

Major neuroleptics...

- Generally the **cis isomer** display more neuroleptic potency than the corresponding trans isomer.
- The structure-activity relationships of the thioxanthenes mimic that of the phenothiazines
- Examples of **thioxanthene** derivatives that are potent



X= Cl, clopenthixol

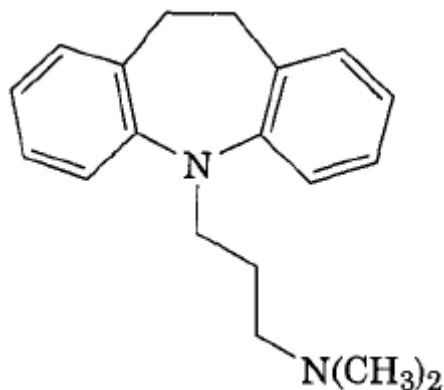
X= CF₃, flupentixol

Major neuroleptics...

- The **phenothiazines** and **thioxanthenes** have the characteristic of two phenyl groups fused to a central six-member ring giving a so-called 6-6-6 system.
- An ethylene is a commonly used bioisostere for a thio linkage in medicinal chemistry
- Continued research in the area of the tricyclics led to the use of this and other two-atom linkages to provide the 6: 7: 6 ring system called dibenzazepine derivatives.

Major neuroleptics...

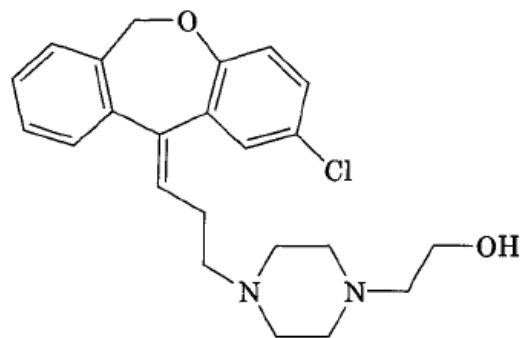
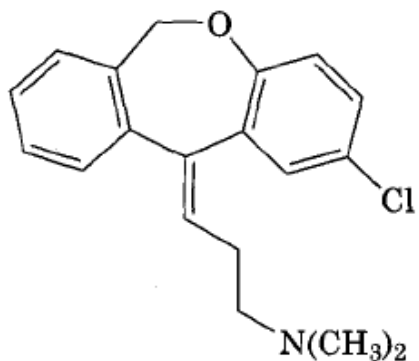
- During clinical studies of dibenzazepine derivatives, it was observed that **imipramine**, unlike the thioxanthenes, was a relatively ineffective antipsychotic agent but seemed to have efficacy in the treatment of **depression**.



imipramine

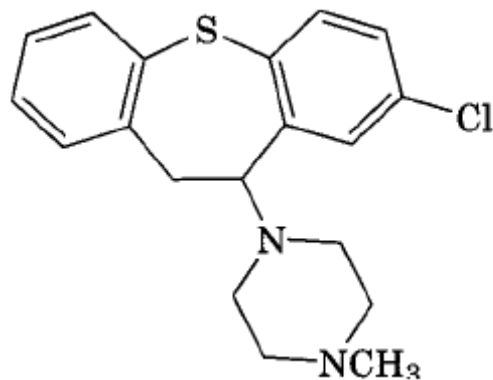
Major neuroleptics...

- Among the various dibenzazepine, two-substituted derivatives had observable neuroleptic actions
 - a) The dibenz[b,e]oxepin derivatives showed chlorpromazine indices of 1 and **6**, respectively



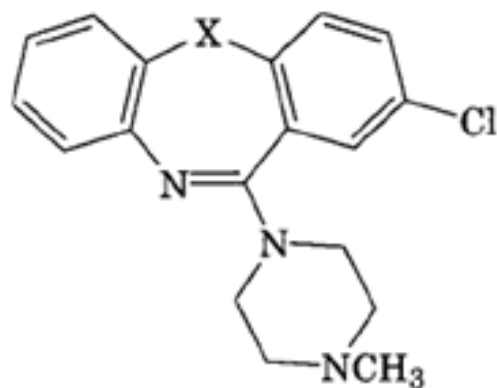
Major neuroleptics...

- b) Dibenz[b,f]thiepins- representative of this class is **octoclothiepine** which is approximately CPZ index =6



Major neuroleptics...

- Another important class of neuroleptics is the dibenzazepines, which are represented graphically in Fig. below.



X = **O** oxyclozapine

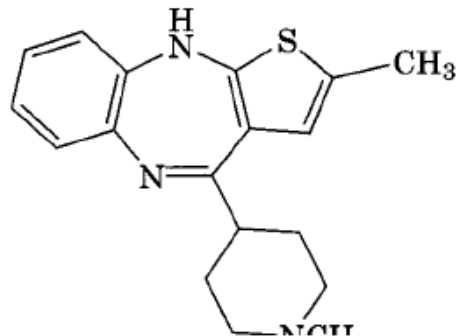
X = S clotiapine

X = NH clozapine

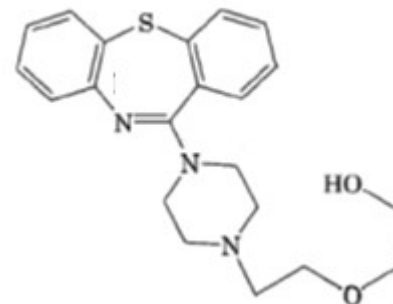
X = CH₂ perlapine

Major neuroleptics...

- From these agents, clozapine does not produce extrapyramidal side effects.
- Two drugs related to clozapine include the almost identical compound, olanzapine, and quetiapine.
- Both confer similar antipsychotic profiles and a minimal propensity to elicit extrapyramidal side effects



Olanzapine



quetiapine

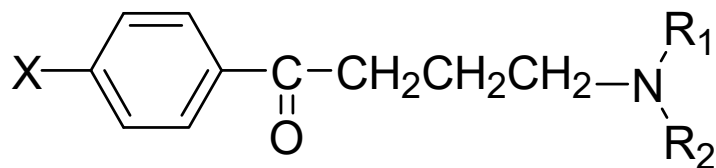
Major neuroleptics...

2. Butyrophenones

- Are chemically unrelated to phenothiazines but have similar activity
- Display high neuroleptic potency

Major neuroleptics...

- **Structure activity relationship**

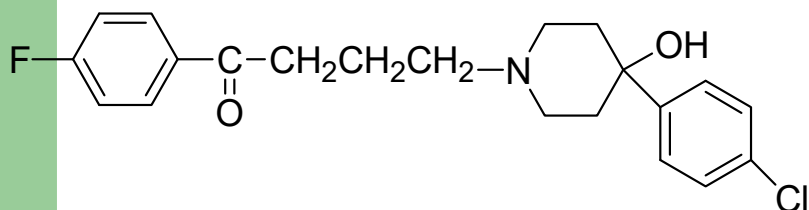


- Fluorine is usually found in the para position (at X) in most potent drugs.
- All butyrophenones possess a tertiary amine at C4 of butyl chain
- Lengthening, shortening or branching of propyl chain decreases neuroleptic potency

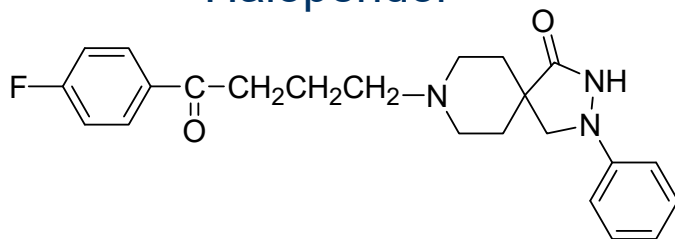
Major neuroleptics...

- The ketone group is important for activity
 - Replacement with thioketone, phenoxy, olefinic or hydroxyl decreases potency
- The tertiary amine may be incorporated into a variety of six membered ring without losing potency
 - Could be piperidine, tetrahydropyridine or piperazine

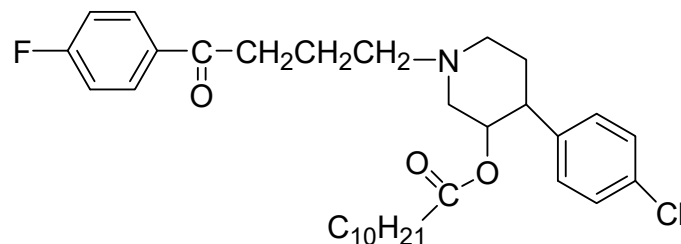
Currently available butyrophenones



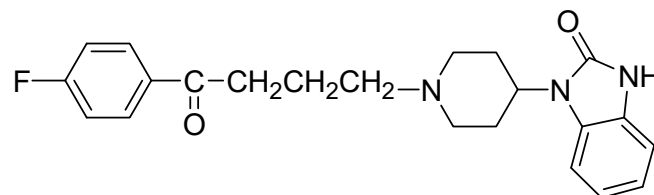
Haloperidol



Spirperone



Haloperidol decanoate



Droperidol

Major neuroleptics...

- **For long term treatment**

- The hydroxyl group of haloperidol provides opportunity to attach alkyl ester group
- Haloperidol decanoate serves as a prodrug releasing haloperidol into the blood stream over a period of time (i.m.)